

Umut Aslan

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Mechatronics Engineer and Master's student in Robotics & Intelligent Systems with a strong foundation in Deep Learning and Artificial Intelligence. Currently driving digitalization projects at BSH Türkiye, bridging the gap between Industry 4.0 concepts and practical factory implementation. Passionate about developing AI-driven solutions for autonomous systems, control systems and embedded systems.

EDUCATION

Master's Degree in Robotics and Intelligent Systems Turkish German University Thesis Subject: Reconstruction of Visual Stimuli from EEG data with Mamba.	Sep 2024 – Present Istanbul
Bachelor's Degree in Mechatronics Engineering Marmara University Dissertation Subject: Welding Quality Control with YOLO-v4	2019 – 2024 Istanbul

WORK EXPERIENCE

Digital Transformation Master Project Student (Global Production / Business Excellence) <i>BSH Türkiye</i>	Aug 2025 – Present Cerkezkoy, Türkiye
- Supporting digital transformation initiatives across production in collaboration with Business and Industry 4.0 teams mainly in Data Analytics and AI solutions within the Dish Care and Cooking production divisions. - Evaluating AI, Data Analytics, and Computer Vision projects based on business impact, feasibility, scalability and ROI. I have performed 'AI Storyboard' initiative for Digitalization Projects. - Contributing to the implementation of digital solutions and AI tools which aimed at improving manufacturing efficiency and quality performance.	
Operational Technologies Intern (Long Term) <i>Toyota Boshoku Türkiye</i>	Feb 2024 – June 2024 Sakarya, Türkiye
- Contribute Digital Transformation initiatives within the industry 4.0 scope, transitioning factory operations from paper-based workflows to digital systems. - Developed and trained a YOLO-based Computer Vision model to automate quality control for detecting mistakes in robotic welding processes. - Utilized Python and Deep Learning techniques to perform visual inspections.	
Research & Development Intern <i>QPFace Information Technologies</i>	Oct 2023 – Feb 2024 Istanbul
- Designed a custom Octocopter drone and integrated specialized sensors for explosive detection. - Conducted R&D on explosive detection systems, designing an odour-sensing mechanism. - Leveraged Fusion 360 software for rapid prototyping and mechanical design of drone components and sensor housings.	
Research & Development Intern <i>Marmara MEMS/NEMS/MOEMS Lab</i>	July 2022 – Sep 2022 Istanbul
- Fabricated MEMS based pressure sensors from Graphene and Carbon-Nanotube. Achieved higher sensitivity and resistance ranges compared to standard industrial sensors. - Implemented data acquisition pipelines using ESP32 and Arduino to integrate sensors into wearable devices.	

TECHNICAL SKILLS and HOBBIES

AI & Computer Vision | Deep Learning, Machine Learning, Image Processing, YOLO, Pytorch

Programming | Python, C, C++, MATLAB, Simulink.

Embedded Systems & PLC | STM32, Arduino, ESP32, Siemens S7-300/400.

Mechanical Design & Analysis | Fusion 360 (CAD/CAM/FEA), SolidWorks.

Languages | Turkish (Native), English (C1).

Parlak Gezegen (Sci-Fi Novel) | Authored and published the science fiction novel 'Parlak Gezegen'.

Chess | Licensed player, Türkiye Chess Federation.